

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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PDB Info	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	JOB ID: 3750189754
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Lipid PDB:	step4_lipid.pdb (view structure)	download.tgz
Assembly Input:	step5_assembly.inp	
Assembly Output:	step5_assembly.out	
System Information:	step5_assembly.str	
Assembled PSF:	step5_assembly.psf	
Assembled CRD:	step5_assembly.crd	
Assembled PDB:	step5_assembly.pdb (view structure)	

Determined System Size:

# of Atoms	77786		
Crystal Type	TETRAGONAL		
System Size	A	94.9137157	Dimension along the A (X) axis
	B	94.9137157	Dimension along the B (Y) axis
	C	94.2717514	Dimension along the C (Z) axis
Crystal Angle	Alpha	90.0	Angle between the axis B and C
	Beta	90.0	Angle between the axis A and C
	Gamma	90.0	Angle between the axis A and B
# of Lipids	on Top	115	
	on Bottom	112	
# of Water	14317		
Z Center	0.0	Center of the system along the Z axis	

Force Field Options:

CHARMM36m ☒

☐ WYF parameter for cation-pi interactions

☐ Hydrogen mass repartitioning

☐ Multi-site Ca²⁺

Y. Gao, J. Lee, I.P.S. Smith, H. Lee, S. Kim, Y. Qi, J.B. Klauda, G. Widmalm, S. Khalid, and W. Im *Journal of Chemical Information and Modeling* 2021 61 (2), 831-839. [DOI](#).

Currently only available for CHARMM, NAMD, GROMACS, and OpenMM
Zhang, A., Yu, H., Liu, C. *et al.* The Ca²⁺ permeation mechanism of the ryanodine receptor revealed by a multi-site ion model. *Nat. Commun.* 11, 922 (2020). [DOI](#).

Input Generation Options:

- ☒ More CHARMM minimization during input generation
- ☐ NAMD
- ☒ GROMACS
- ☒ AMBER
- ☒ OpenMM
- ☒ CHARMM/OpenMM
- ☐ GENESIS
- ☐ Desmond
- ☐ LAMMPS
- ☐ Tinker

Equilibration Options:

- ☐ P21 image transformation (only available for CHARMM)
- ☐ CHARMM DOMDEC (only available for CHARMM)
- ☒ Generate grid information for PME FFT automatically
- ☐ Explicit grid information for PME FFT

X	Y	Z

- ☐ NVT ensemble
- ☒ NPT ensemble
- ☐ NPAT ensemble
- ☐ NPgT ensemble

Surface Tension (dyne/cm)

Temperature: K

Next Step: 
Generate Equilibration and Dynamics Inputs



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